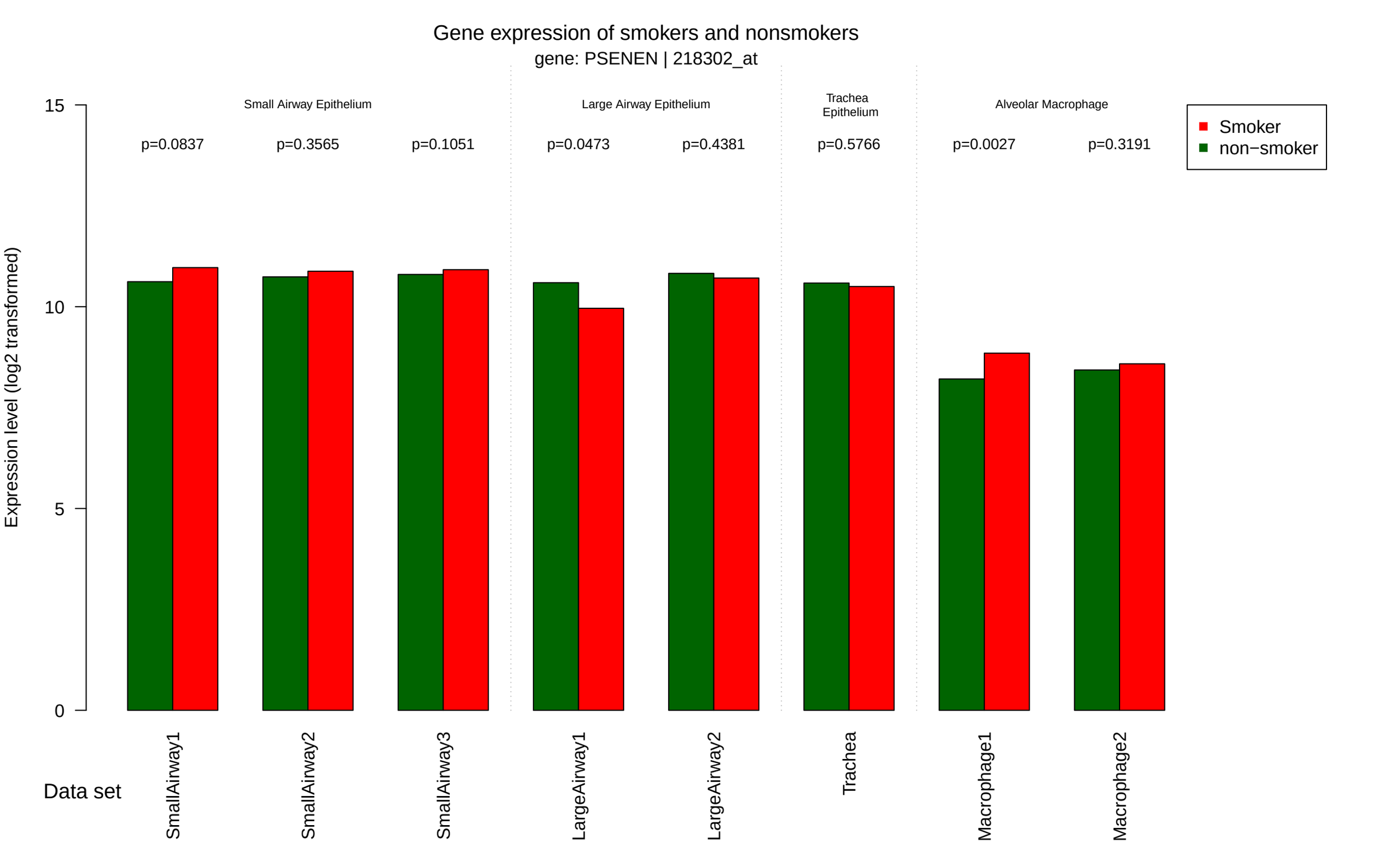
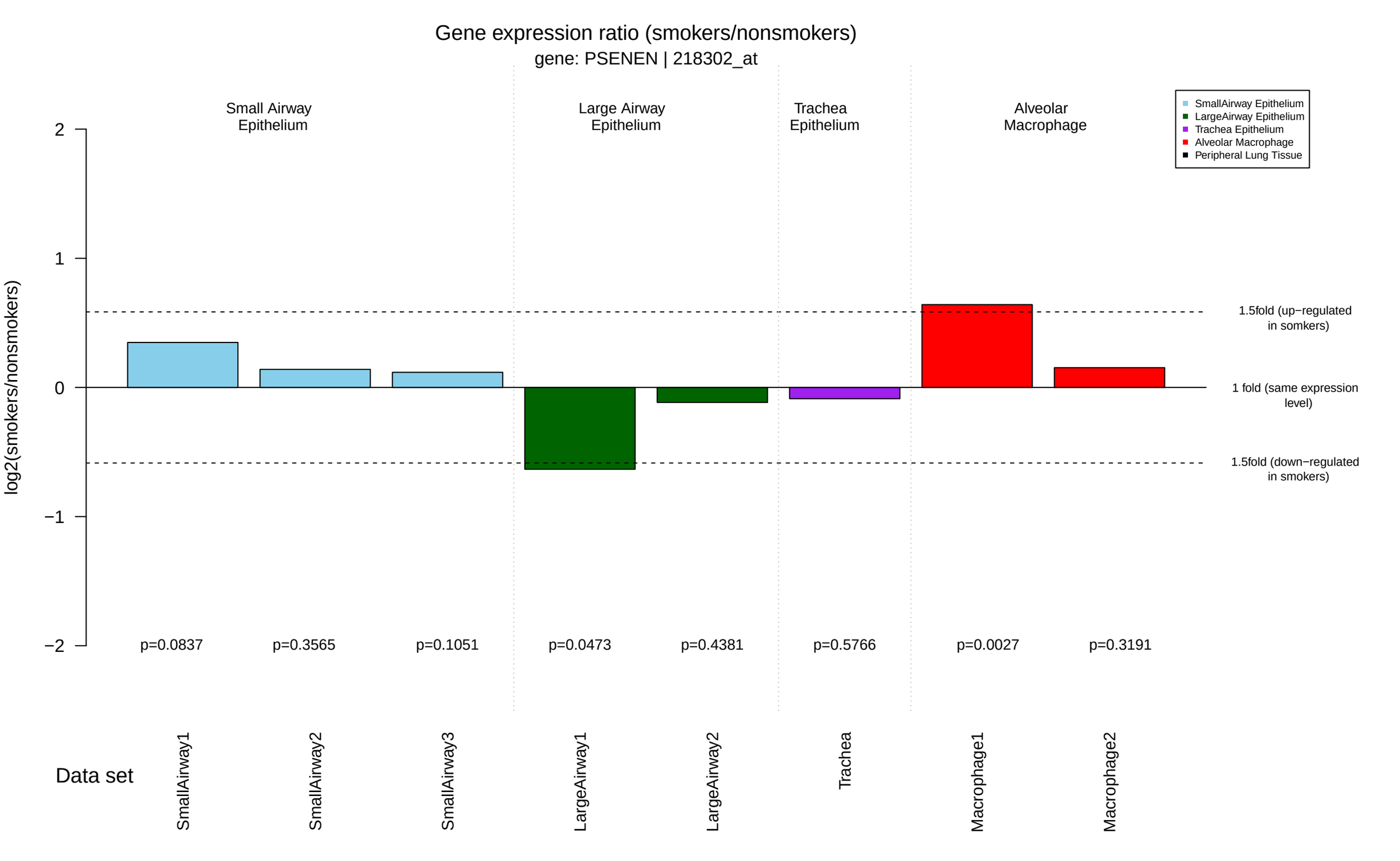
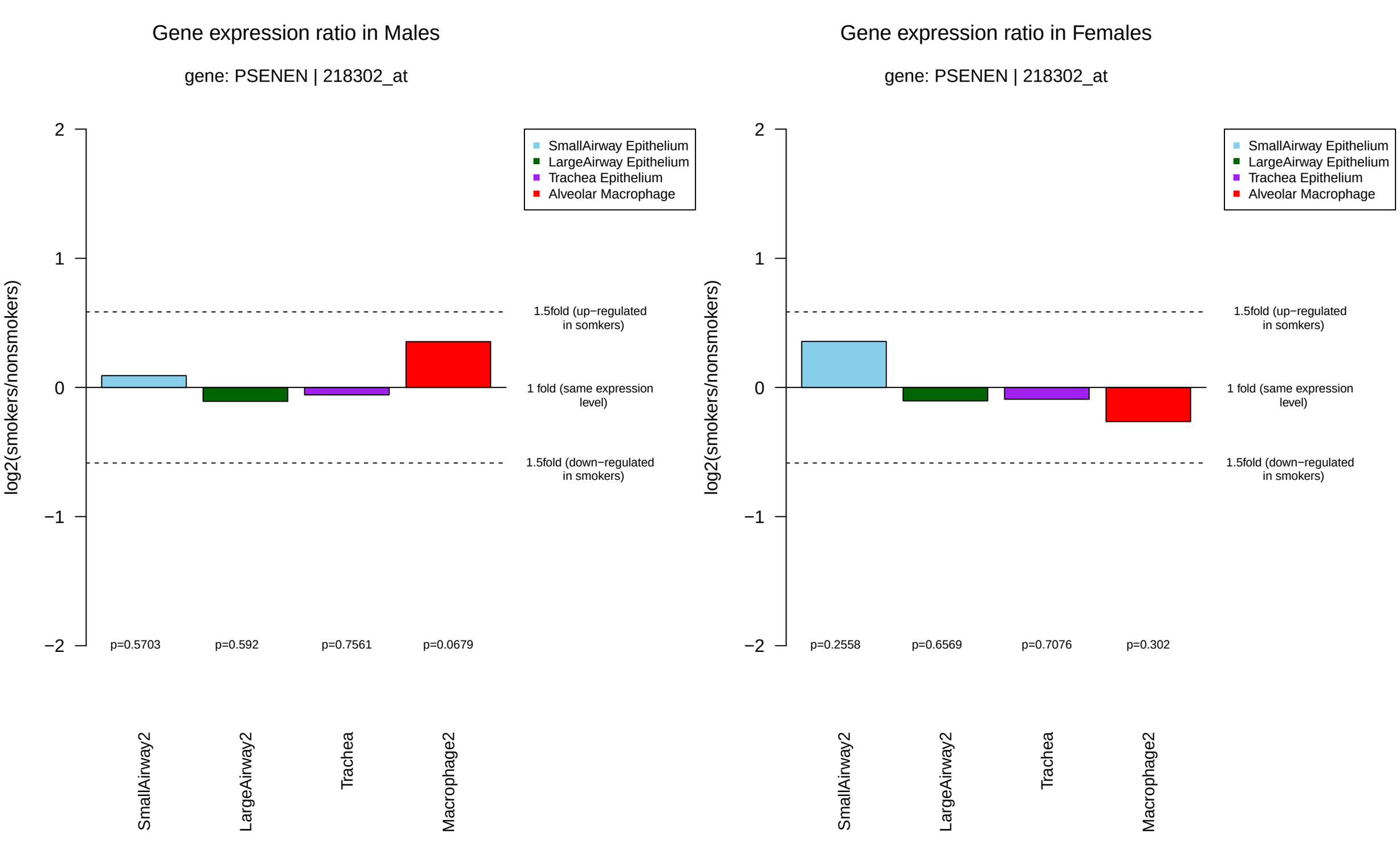
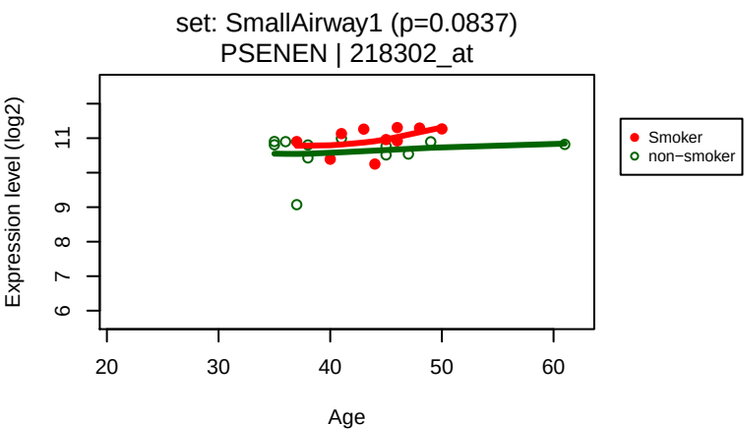
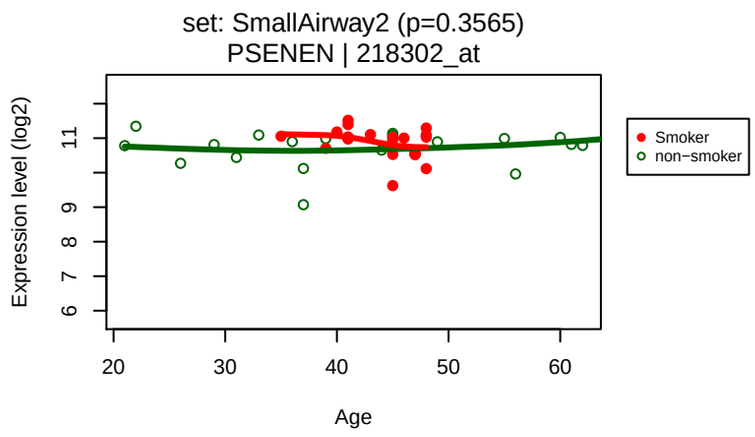
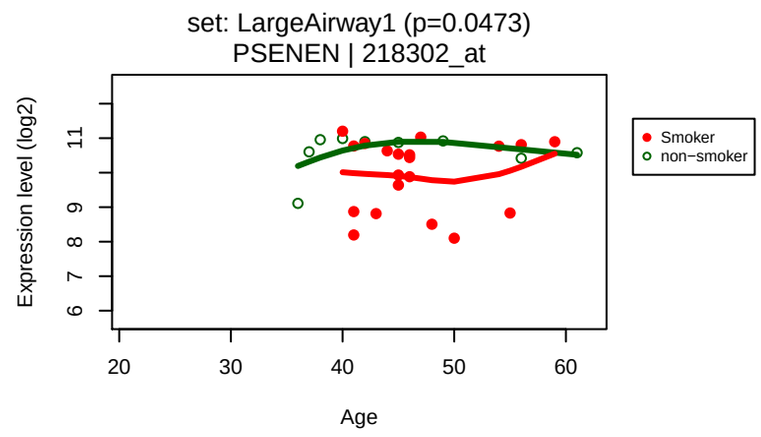

Gene expression of smokers and nonsmokers
gene: PSENEN | 218302_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0837
p=0.3565
p=0.1051
p=0.0473
p=0.4381
p=0.5766
p=0.0027
p=0.3191
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

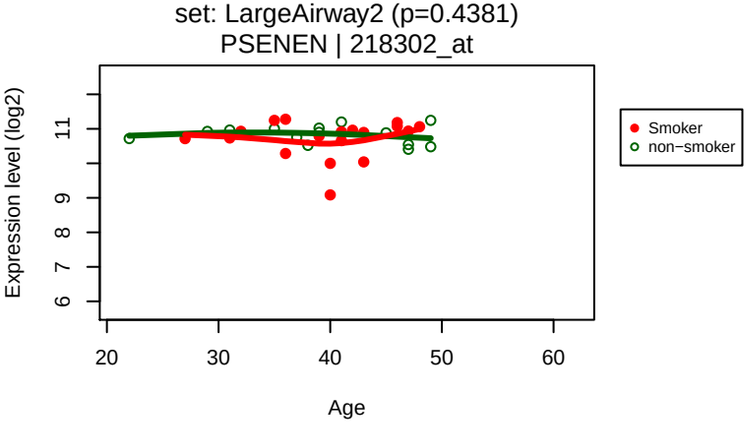

Gene expression ratio (smokers/nonsmokers)
gene: PSENEN | 218302_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0837
p=0.3565
p=0.1051
p=0.0473
p=0.4381
p=0.5766
p=0.0027
p=0.3191
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

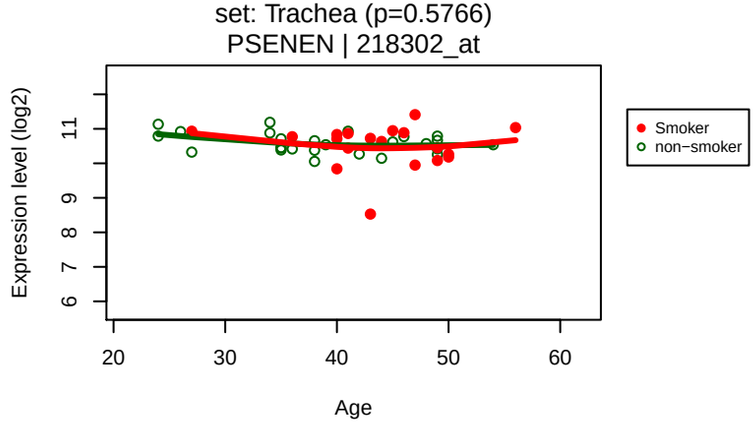

Gene expression ratio in Males
gene: PSENEN | 218302_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5703
p=0.592
p=0.7561
p=0.0679
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSENEN | 218302_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2558
p=0.6569
p=0.7076
p=0.302
SmallAirway2
LargeAirway2
Trachea
Macrophage2

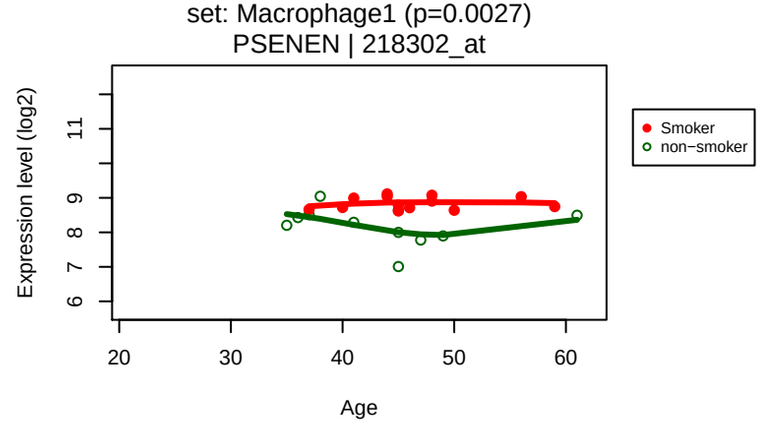

set: SmallAirway1 (p=0.0837)
PSENEN | 218302_at
Expression level (log2)
Age
Smoker
non−smoker

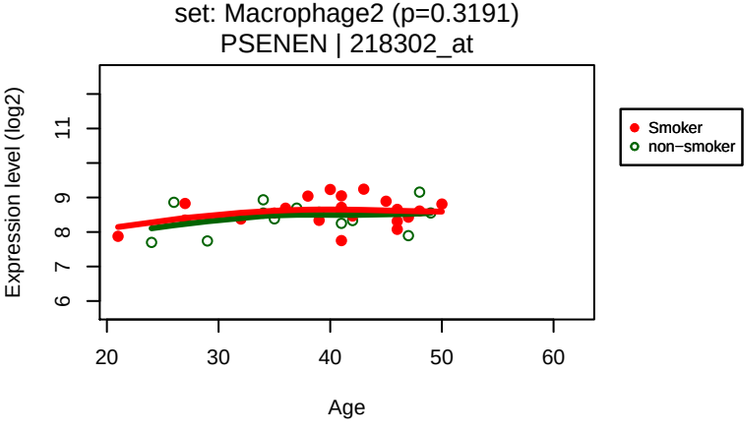

set: SmallAirway2 (p=0.3565)
PSENEN | 218302_at
Expression level (log2)
Age
Smoker
non−smoker

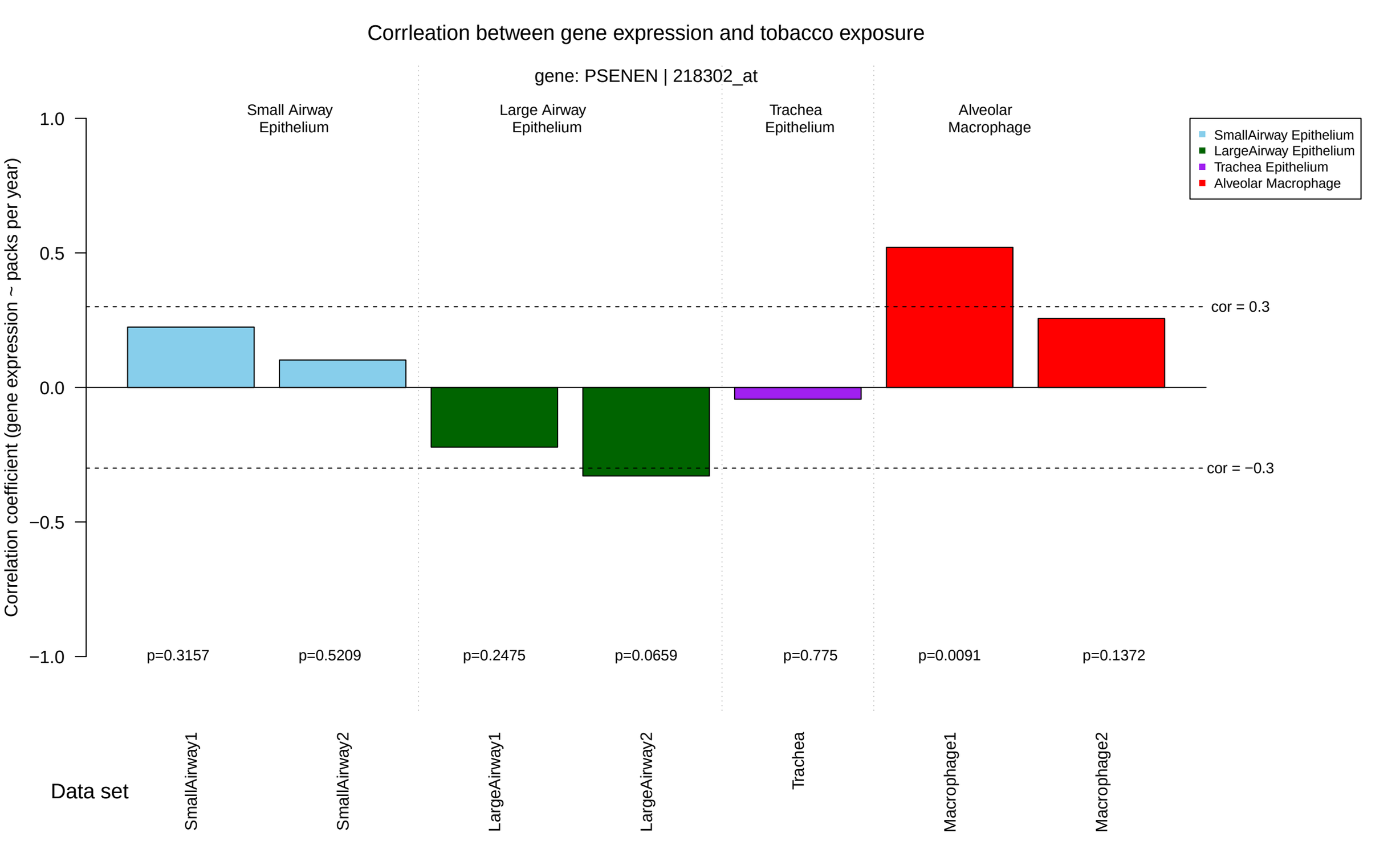

set: LargeAirway1 (p=0.0473)
PSENEN | 218302_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4381)
PSENEN | 218302_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5766)
PSENEN | 218302_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0027)
PSENEN | 218302_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3191)
PSENEN | 218302_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSENEN | 218302_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3157
p=0.5209
p=0.2475
p=0.0659
p=0.775
p=0.0091
p=0.1372
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2